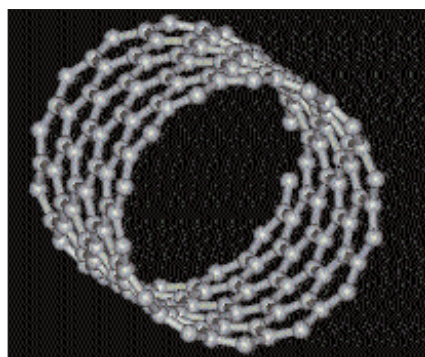




An illustration showing the atomic structure of a carbon nanotube

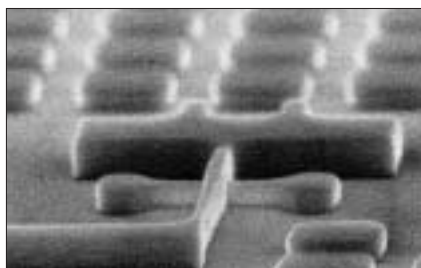
the two come in contact and resistance is lowered. This is considered as 'one'. Thus, ones and zeros are stored in the memory. These miniscule nanotubes provide for a lot of storage space within a small device. The company is hoping to bring out a working prototype by early 2005. Intel is also researching carbon nanotubes.

Three-dimensional transistors

Tri-gate transistors are non-planar and have a 3D structure. The current transistors have a planar structure—the current flows on a single plane. However, a tri-gate transistor is like a building in which current flows vertically as well as horizontally. Speaking about the limitations of the planar design, Dr. Gerald Marcyk, director of the Components Research Lab, Intel, says, "Our research shows that below 30 nanometers, the basic physics of the flat, single-gate planar transistor leaks too much power to meet our future performance goals".

In a tri-gate transistor, current flows on the top surface and also on the vertical walls. This gives more space for current flow, which, in turn, results in lower current leakage. Thus, chips based on tri-gate transistors can be made to run faster than planar chips by about twenty per cent. In 2001, Intel introduced a new class of transistors called TeraHertz transistors. These can switch between the binary states at a speed of 1000 GHz. The TeraHertz transistor is the result of three developments, namely, high-k/metal gate, tri-gate and finally, the burying of an oxide layer within the silicon substrate. This layer of oxide blocks unwanted flow of current, thereby reducing current leakage. Intel plans to commence the production of TeraHertz transistors before the end of this decade.

As the name indicates, a tri-gate transistor has three gates. The gate



Magnified image of the microscopic Intel tri-gate transistor as seen through an electron microscope

controls the flow of current from the source to the drain in a transistor. Use of multiple gates increases the current flow. IBM and AMD are also working on multi-gate transistors. In September 2002, AMD claimed that it had developed a double-gate transistor. A year later, it announced that with the help of IBM, it plans to use tri-gate transistors in its chips. Like other innovations, this technique is also compatible with existing manufacturing facilities. "This multi-gate transistor implementation is highly compatible with current manufacturing techniques, which improves our ability to put this technology into high-volume production," says Craig Sander, vice-president of process technology development for AMD.

New lithography techniques

Currently, optical lithography is used to manufacture chips. A combination of a light beam, mask and photo-resist is used to build layers on substrate's surface and etch transistors and their connections. As the size of the transistors keeps getting smaller, the lithography process should be modified. Deep Ultra Violet Lithography (DUVL) employs Deep Ultra Violet (DUV) beams with a wavelength of 248 nm. DUVL allows manufacturers to print circuits as narrow as 0.1 μm . To print circuits smaller than that, manu-

facturers will have to look for alternatives. Extreme Ultra Violet (EUV) lithography seems a likely successor to DUV. It's possible to print circuits up to 0.03 μm using EUV, since its wavelength is around 13 nm. Although they might sound similar, DUV and EUV technologies have a lot of difference between them. This is because the EUV portion of the electromagnetic spectrum is vastly different from the UV and visible light spectrum. One of the challenges posed by EUV Lithography (EUVL) is that the optics used in the manufacture should have unprecedented levels of accuracy. There are many technical hurdles like these that have to be crossed for it to be implemented.

There are other lithography techniques such as X-ray lithography, ion beam projection lithography, and immersion lithography that may come in EUVL's way. Immersion lithography shows a lot of promise. Its principle is to improve the resolution of optical lithography by using a thin layer of pure water. If this is successful, then it may make EUVL redundant, or delay its implementation.

Joint winners

Two observations can be made from what has been explained. One is that all these techniques will complement each other; there aren't many competing with one another. The second observation is that most of the developments have been made either by Intel or IBM, thanks to the billions of dollars poured into R&D by these companies. Other companies can't afford such huge bills and so, in the future, they will find it prohibitive even to build manufacturing plants. The not-so-famous Moore's Second Law states that the cost of building a new manufacturing plant will double every four years. Indeed, those costs have been spiralling and will continue to do so.

Here, a special mention should be given to other forms of computing such as quantum computing and biological computing. Although they are still in their infancy, they hold promise. If any of them fulfil what they predict, then we may probably have to dump all the current technology, but until then, the semiconductor industry will continue to travel at the rate demanded by Moore's Law. ■

MOULY ARUN PRABHU

mouly_arunprabhu@thinkdigit.com



www.research.ibm.com

Apart from silicon germanium technology and VLSI design automation, this Web site is crammed with information about storage, data management, and leading-edge application research that IBM is doing.

www.intel.com/research

This is another Web site that goes into the nitty gritty of nanotechnology and silicon. It also has tickers on Intel news and upcoming events.